

Homework #1:

Due date: Sep. 21st 2025

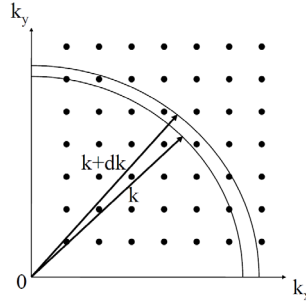
1. Two-charge-carrier Drude model:

Consider a system of two types of charge carriers in the Drude model. The two carriers have the same density (n) and opposite charge (e and $-e$), and their masses and relaxation rates are m_1 , m_2 and τ_1 , τ_2 , respectively. So their corresponding mobilities are $\mu_1 = \frac{e\tau_1}{m_1}$ and $\mu_2 = \frac{e\tau_2}{m_2}$, respectively.

- (a) Calculate the magnetoresistance $\Delta\rho = \rho(B) - \rho(B = 0)$, where B is the magnetic field.
- (b) Calculate the Hall coefficient.
- (c) In an undoped semiconductor, $n = n_0 e^{-\Delta/k_B T}$ describes the temperature dependence of the carrier concentration. What will the temperature dependence of the magnetoresistance and the Hall coefficient be?

2. Density of States in 2D system:

Consider free electron in a 2D box with a width of a . The allowed wave-numbers k_x and k_y for x and y directions respectively are expressed as $\frac{\pi n_x}{a}$ and $\frac{\pi n_y}{a}$, where $n_x, n_y = 1, 2, 3, \dots$ are integer numbers. The following figure shows the schematic view of the 2D array of allowed quantum states in k -space.



- (a) Determine the area in k -space occupied by one allowed k point.
- (b) Express number of allowed states in the area enclosed by the first quarter circles with the radius of k and $k + dk$, where $k \gg dk$. (Consider the spin degeneracy)
- (c) Express density of allowed states in real space per unit energy.

3. Density of states from a general formula

- (a) In a 3D solid and starting from the general formula, $g(E) = \frac{2V}{(2\pi)^3} \iint \frac{dS_E}{|\nabla_k E|}$, find the expression for the electronic density of states: (i) for free electrons; (ii) for electrons that obey the isotropic dispersion relation: $E = -E_0 - 2\gamma \cos(\frac{ka}{2})$, where E_0 , $\gamma > 0$.
- (b) After having established the corresponding expression for 2D, find $g(E)$ for the hypothesis (i) and (ii) above.
- (c) Same question for 1D.